UNIVERSITY OF LONDON

BUSINESS ANALYSIS REPORT

Tableau Visualizations

Prepared for
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Senior Management

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Executive Summary

With the provided dataset, an analyst will be using Tableau's visualization tools to analyze trends and relationships that might be helpful for increasing sales.

The following lists the visualization tools that we have used in Tableau to support the analysis:

1. Stacked Bar
2. Area Chart
3. Bar Chart
4. Maps
5. Packed Bubbles
6. Line Chart
7. Scatter Plot
8. Highlighted Table

For the story, an analyst has already created interactive dashboard that allow you to view the relationship by dragging the cursor over the section you want to view.

1. Introduction

1.1 Background

As business analyst, with provided sales data of four years (from 2019 until 2022) with 51,229 rows of value.

1.2 Scope

The following report analyze the sales data to provide key insights and the optimal idea for senior management.

2. Methodology

2.1 Purpose of Report

The purpose of the report is to successfully demonstrate the business growth over the last four years and to encourage senior management to make a decision based on the factual data of the trends in this report.

2.2 Approach

The analyst will use variety of Tableau visualizations to identify any significant trends or relationship between the qualitative and quantitative data.

3. Dashboard Analysis

3.1 Comprehensive Sales Insights

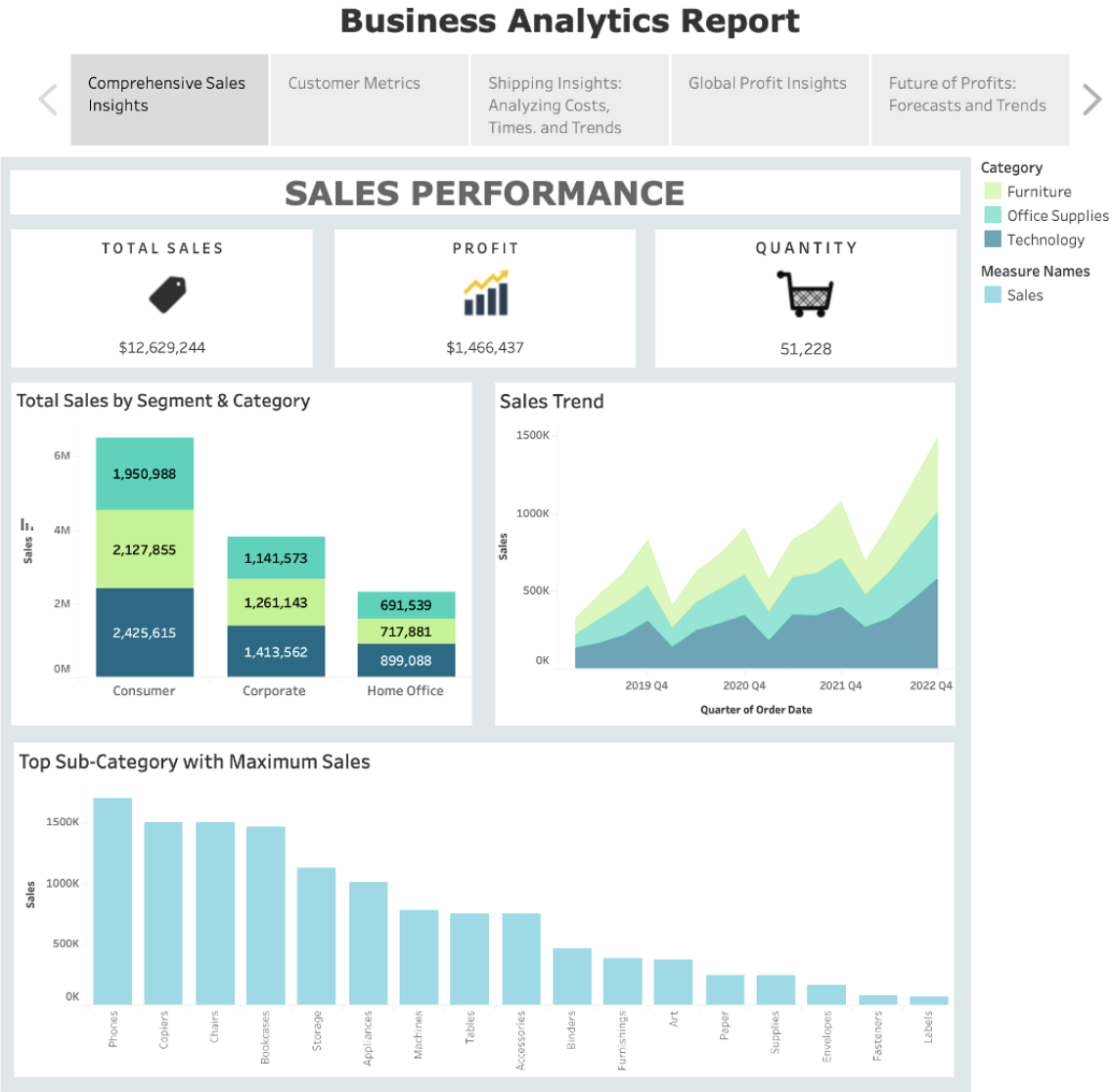

Figure 1: Dashboard 1 (Comprehensive Sales Insights)

Observations:

1. Stacked Bars – Total Sales by Segment and Category

   Consumer is the highest contributor to the sales across the categories, with technology leading sales in other segments as well. The sales of home office, however, is significantly distinct from consumer sales which account for roughly one-third of consumer sales, suggesting that the company's marketing and product development may be more in line with consumer needs than that of corporate and home office.

2. Area Charts – Sales Trend

   Overall, the category with the largest sales is still technology in every quarter. The company is undergoing sales growth as all categories are increasing over four quarters starting from Q4 2019 to Q4 2022. There is seasonality in sales as it tends to culminate in the fourth quarter of the year, corresponding with the holiday seasons which typically related to higher consumer spending.

3. Bar Charts – Top Sub-Category with Maximum Sales

   Phones are the top-selling throughout the sub-category as technology dominates the sales across the category. The sales performance from phone to labels witnessing a decrement, which indicates wide range of performance, suggesting that some might be underperforming.

Recommendations:

Given the robust sales in the consumer segment and technology category, more specifically in phones, merge these insight into creating packaged offers, targeting the consumers with technological demands particularly during the fourth quarter peak sales period to maximize revenue especially during peak season and leverage consumer behaviour who has preferences in technology. Moreover, in order to mitigate the lower sales volumes recorded in the corporate and home office segments, create a targeted campaign during the off-peak seasons.

3.2 Customer Metrics

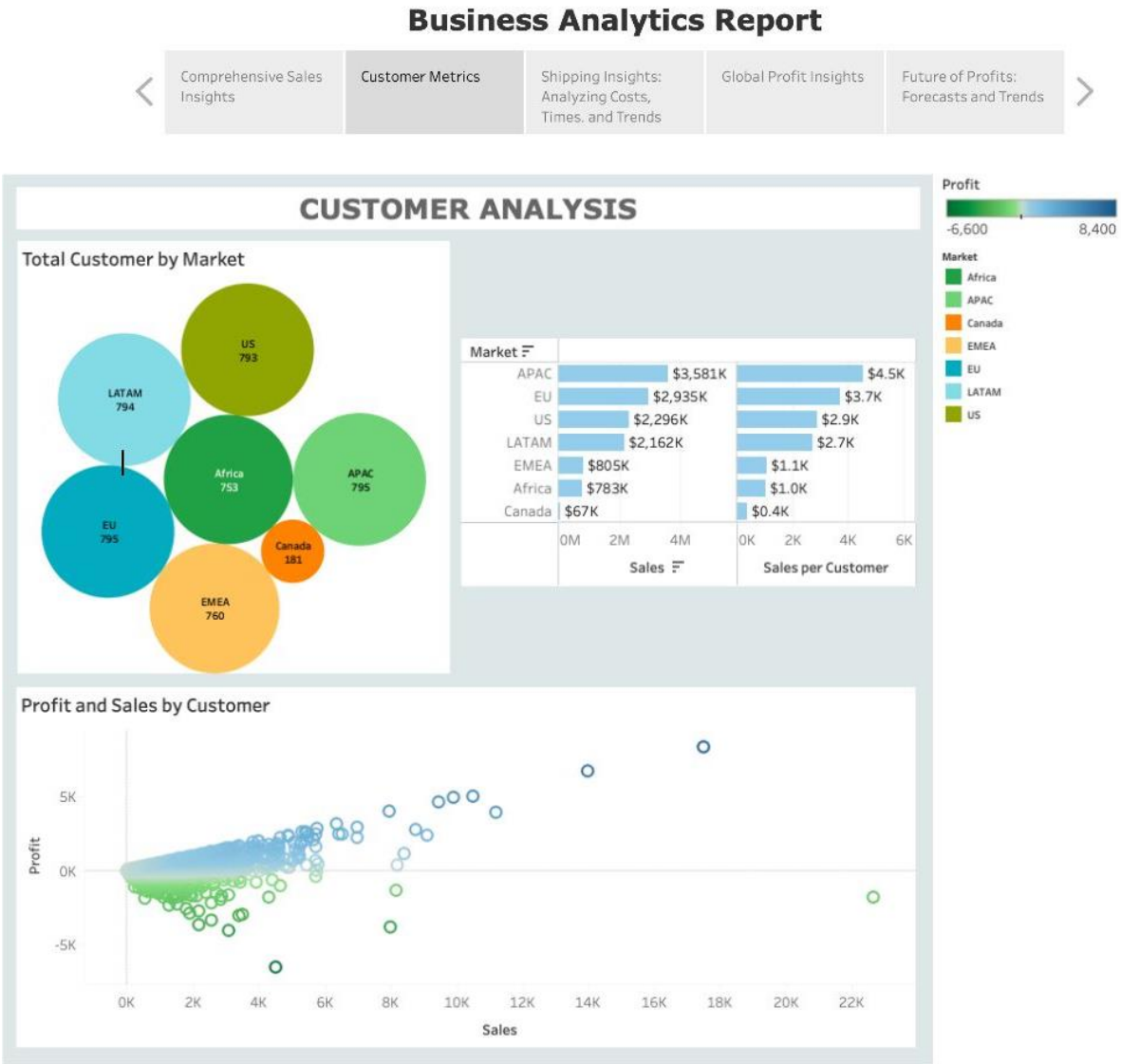

Figure 2: Dashboard 2 (Consumer Metrics)

Observations:

1. Packed Bubbles – Total Customer by Market

   The size of the circles indicates the number of customers in each market. The customers distribution is even throughout the 6 markets, all slightly under 800 while Canada having the fewest.

2. Bar Charts – Sales and Sales per Customer by Market

   Despite the same number of customers in APAC and the EU, APAC's sales and sales per customer are higher than those in EU. The number of customer in Africa is not proportional to the sales and sales per customer.

3. Scatter Plot – Profit and Sales by Customer

There is a positive correlation between sales and profit as sales increase as profit increases. However, there are outliers with high sales but has negative profit, indicating high sales do not always equivalent to high profitability.

Recommendations:

Perform a thorough analysis of customer behaviour and adjust the sales strategy to figure out why sales and sales per customer are higher in APAC according to the same number of customers with the EU. Utilizing cross-market strategies to increase customer retention while improving customer acquisition in underpenetrated markets like Canada. Furthermore, make sure that sales efforts are generating both profitability and revenue.

3.3 Shipping Insights: Analyzing Costs, Times, and Trends

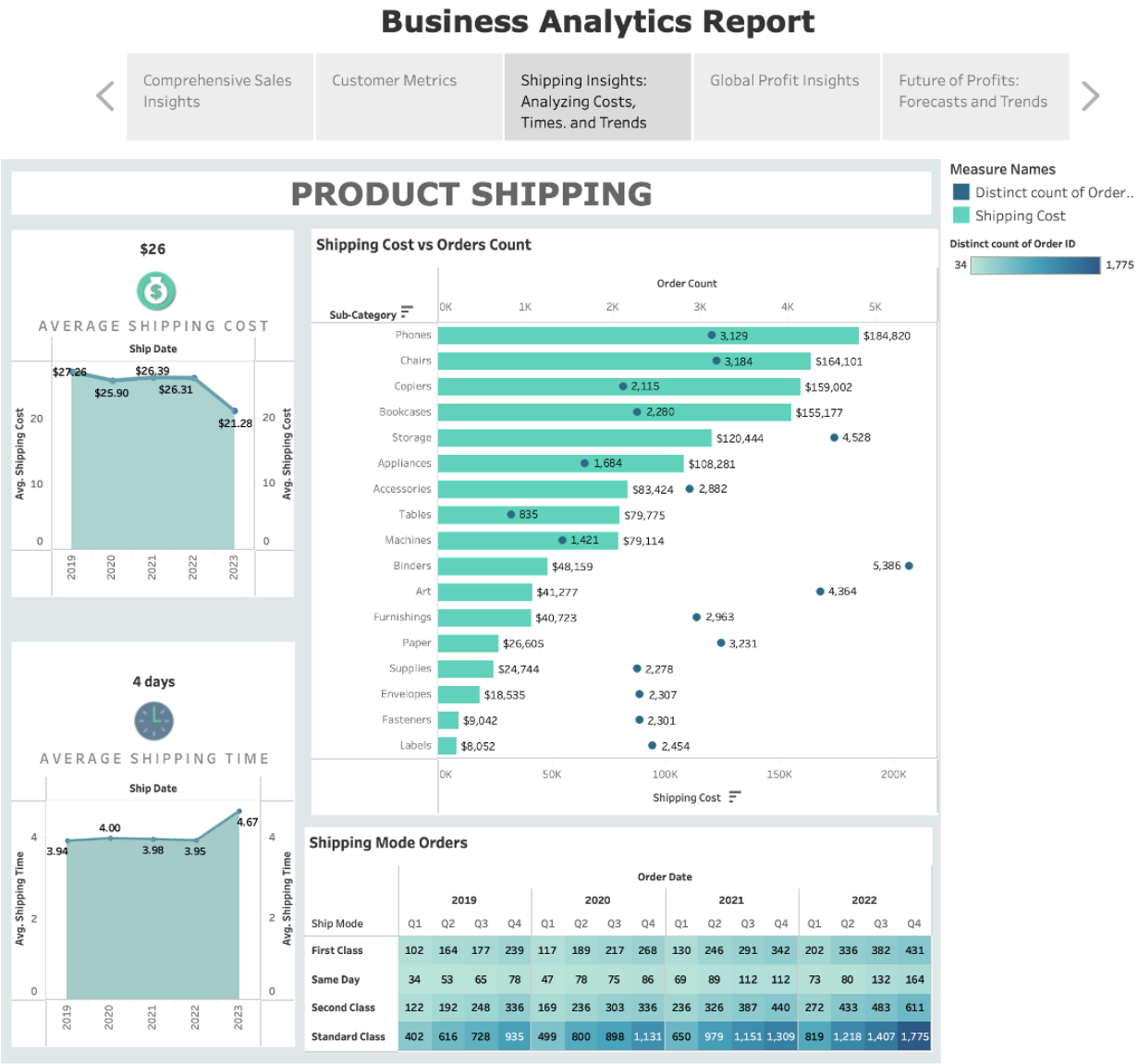

Figure 3: Dashboard 3 (Shipping Insights: Analysing Costs, Times, and Trends)

Observations:

1. Line Charts – Average Shipping Cost and Average Shipping Time

   The average shipping cost has decreased over the years from $27.26 in 2019 to $21.28 in 2023. The average shipping time has slightly increased from 3.94 days in 2019 to 4.67 days in 2023. The increment in the shipping time might be due to the drop in the shipping cost.

2. Bar Charts – Shipping Cost vs Order Counts

   Higher volume categories incur higher shipping costs, which could be due to the size or weight of the items requiring higher cost shipping options.

3. Highlighted Tables – Shipping Mode Orders

   Standard class is the most popular shipping mode with a steady growth in usage over the years. Customers tend to prefer low cost shipping mode over speed, as demonstrated by the popularity of standard class shipping.

Recommendations:

A cost-benefit analysis need to be conducted in order to figure out whether the benefits of faster shipping modes, such same day, worth the costs. In order to discover more regarding the shipping trends and pinpoint areas for costs savings, implement a supply chain analytics system.

3.4 Global Profit Insights

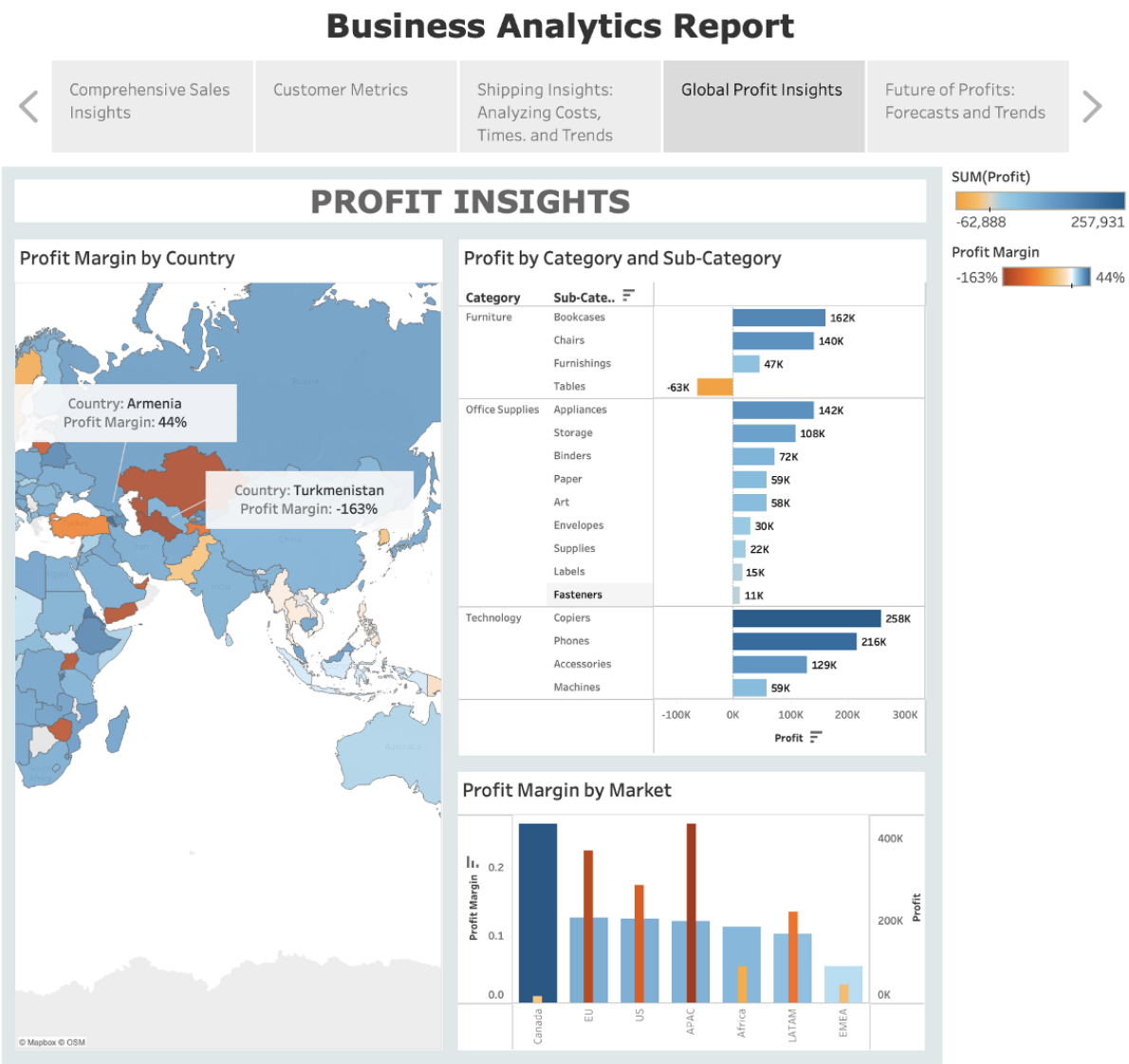

Figure 4: Dashboard 4 (Global Profit Insights)

Observations:

1. Maps – Profit Margin by Country

   Armenia is making a healthy profit margin of 44%, showing strong market demand or effective operations. On the other hand, Turkmenistan is generating a profit margin of -163%, indicating that the cost surpass the revenue.

2. Bar Charts – Profit Margin by Market

   APAC has both high profits and profit margins. EMEA has the lowest profits and the second lowest profit margin. Canada has high profits yet low profit margins. The significance difference between the markets indicating diverse market dynamic.

3. Bar charts - Profit by Category and Sub-Category

   Technology yields the highest profit with copiers generate the most, whilst tables form furniture category is the only product witnessing loss. The Copiers' high profit showing there is a strong market demand and efficient strategy, while the loss by tables might be the issues arising from tight competition or cost.

Recommendations:

Ensuring the pricing strategy like cost reduction and price optimization aligns with cost structure and market conditions to meet the specific requirements and level of competition in each markets in order to manage profit margins. Moreover, identify where to focus on advertising and sales by evaluating the performance of products not only inside categories but also across markets.

3.5 Future of Profits: Forecast and Trends

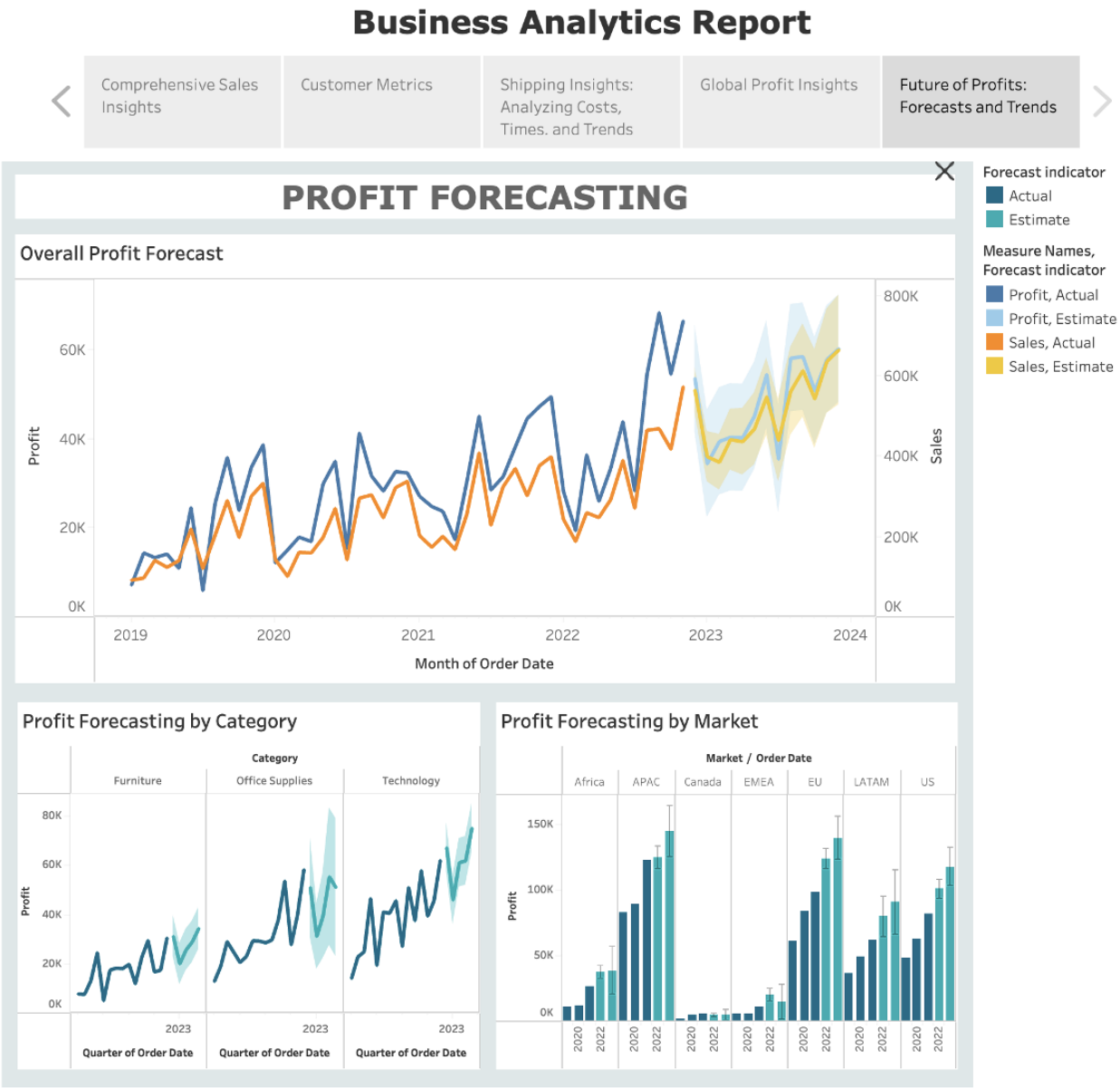

Figure 5 : Dashboard 5 (Future of Profits: Forecasts and Trends)

Observations:

1. Line Charts – Overall Profit Forecast

   There is a correlation between profit and sales as both metrics are trending upward overall, indicating growth.

2. Line Charts – Profit Forecasting by Category

   Technology has a significant upward trending, while furniture and office supplies depicts a modest growth. It might be due to the increase in demand or higher margins of technology.

3. Bar Charts – Profit Forecasting by Market

   APAC is the most profitable market, followed by the EU and US that might be due to the higher sales volumes or margins. Showing consistent upward trend over the years, suggesting to expand the global demand.

   Recommendations:

   Technology is the main driver of profit growth. Therefore, spread the technology investment across markets, particularly in high-growth market like APAC, in order to take benefit from the technology category's increasing sales trend.

4. Conclusion

In conclusion, maximizing revenue and dealing with sales volume disparities in distinct segments can be obtained by leveraging strong sales in the consumer and technology, particularly in phones, using targeted package offers during peak seasons like in the fourth quarter and other particular campaigns in off-peak times. Analyzing customer behavior across market like APAC and the EU market, paired with cross-market strategies, will improve both customer retention and acquisition, particularly in markets like Canada. In order to boost total profit expansion, it is suggested to concentrate in technology investment especially in high-growing market like APAC as it will assist in leveraging the potential for growth of the technology sector.

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